

Norwich: air quality and noise from official data

Official figures from DEFRA's published pollution and noise maps, compared with World Health Organization guidelines.

0.9×

NO₂ background level against the WHO guideline (9.4 vs 10 µg/m³)

45th

highest air pollution of the 97 UK council areas compared: Norwich (1.49 times the WHO guideline on its worst pollutant)

49%

of Norwich's postcodes are above the WHO road-noise guideline (56th highest of 93)

Air quality, road noise and the airport across Norwich

Location	Postcode	NO ₂ (µg/m ³)	PM2.5 (µg/m ³)	Road noise (dB Lden)	Nearest runway
City centre (NR2)	NR2 1NE	11.1	7.6	53.0	5.3 km north
Norwich station (NR1)	NR1 1EF	12.1	7.5	61.9	5.5 km north
Nelson (NR2)	NR2 3DQ	11.1	7.6	50.0	5.2 km north
Sewell (NR3)	NR3 3HR	9.4	7.6	55.9	3.6 km north
Eaton (NR4)	NR4 7LB	7.9	7.2	51.2	7.3 km north
Bowthorpe (NR5)	NR5 9DS	7.9	7.3	53.6	4.8 km north-east
Catton Grove (NR6)	NR6 6AD	9.3	7.4	48.3	1.5 km north
Crome (NR7)	NR7 9TA	10.0	7.5	52.4	4.4 km north-west
WHO guideline		10	5	53	

Figures in orange exceed the WHO guideline. The UK legal limit for NO₂ is 40 µg/m³, so levels below it are within the law even where they are above what the WHO recommends for health.

What this shows

- **Fine particles (PM2.5) are above the WHO guideline at every point sampled** (7.2 to 7.6 against a guideline of 5). Nitrogen dioxide is above its guideline at 3 of 8 points.
- **Road noise varies street by street far more than air does.** Norwich station (NR1) reaches 62 dB Lden; Catton Grove (NR6) is 48.
- Across the council as a whole, Norwich ranks **45th of 97** UK council areas for air pollution against WHO guidelines, highest first, and 56th of 93 for the share of postcodes above the road-noise guideline. 0.43% of its postcodes are at medium or high flood risk.
- **Norwich Airport's runway is 1.5 to 7.3 km from the points sampled; the nearest is Catton Grove (NR6).** None of them is within 2 km of a published approach line. The government's noise mapping does not cover Norwich Airport, so there is no official aircraft noise figure for anywhere in this area.

Note: DEFRA's air figures are modelled background levels on a 1 km grid (2024 annual means), so they understate pollution at the kerb of main roads, where roadside monitors and sensors take direct measurements. The noise maps model 2021 traffic, reduced by lockdown. Runway distances are measured to the runway itself, from its published position: they are distances, not noise levels. Helicopters, military flights and smaller airfields are not covered.

Sources: DEFRA background pollution maps (PCM), 2024 annual mean, 1 km grid; DEFRA Strategic Noise Mapping Round 4 (road, Lden, 2021 traffic); WHO Global Air Quality Guidelines (2021) and Environmental Noise Guidelines for the European Region (2018). All Open Government Licence v3.0 except the WHO guidelines. Runway and approach positions from the UK Aeronautical Information Publication (NATS), AIRAC 2026-10-01. Council-area comparison across the 97 UK council areas in Sky Score's open data (skyscore.co.uk/open-data). Compiled with Sky Score (skyscore.co.uk), which joins these public datasets; free to reuse and cite. Prepared 2 October 2026.